

Holly Metzger

Honors 290

KT, PhD

19 April 2026

Water World Magnetic Fields: Plausibility and Consequences

Water worlds are exoplanets of size between Earth and Neptune that consist of up to 50 mass percent water. They orbit within their star's habitable zone and have a liquid ocean that covers their entire surface. Their existence is still debated, as there is currently no way to differentiate between a water world and a gas dwarf. One unexplored disambiguation method is the presence of a magnetic field. The measurement of magnetic fields has just become technologically feasible, and when combined with data obtained by other methods, a clearer picture of the interior composition of an exoplanet is possible than ever before achieved.

Among the population of over 6,000 confirmed exoplanets is a radius valley, where there are few planets with radii between 1.5 and 2 times the radius of Earth (Fulton et al. 1). Planets around this size are divided into two classes: super-Earths and sub-Neptunes, depending on which side they fall. Explaining this valley is the subject of much research. Of particular interest is how planets are formed. The bimodal distribution of radii implies that there are two distinct types of planets, which means they either form differently from the start, or undergo some process that changes one of them. Evidence for these populations having distinct chemical compositions would support the hypothesis that they have different formation pathways. However, current methods cannot achieve this, leaving composition as an open question. Observing a magnetic field would put constraints on their composition and evolution, giving researchers a better estimate of their true nature.

The prevailing theory to explain the radius valley says that both classes of planets fundamentally form in the same way. The larger planets above a tipping point of mass can retain

their gaseous atmospheres, resulting in a higher radius. Planets below that point do not have sufficient gravity to retain their atmospheres, which escape and leave only the smaller rocky core (Rogers et al. 1). A parallel theory that complicates this model is water worlds. First proposed by Leger et al in 2004, “ocean planets,” also known as water worlds, or hycean planets, are planets that form outside the frost line, where volatiles such as water exist as solid ice in the protoplanetary disk, then migrate inward so the ice melts. The ice is incorporated into young planets, resulting in a high water mass percent, typically estimated to be between 20% and 50% (Barros, S. C. C. et al. 11). To accurately model the development of this proposed exoplanet population, researchers must account for the effects of a magnetic field.

Water worlds have a smaller metal core and silicate mantle compared to gas dwarfs, made up for by thick layers of ice and liquid water such that their bulk densities are equal. Because bulk density can be measured, but the actual chemical composition cannot, there are “many theoretical models” that have “redundant outputs in terms of the bulk properties” (Carroll 59:16). This degeneracy motivates the search for additional observational constraints, such as magnetic fields, that could provide greater insight into planetary compositions. Magnetic fields are complex and depend on a variety of factors, many of which are currently unknown even for Earth. However, it is precisely because getting a clear picture of a planetary magnetic field requires “understanding of the history and material properties of planets” that researchers could also use observations of a magnetic field to disambiguate those properties (Stevenson 652).

Because the primary determinant for a planetary magnetic field is the presence of a dynamo in its core, determining if a water world would meet the requirements to host one is the first step in assessing its magnetic field probability. The three requirements for sustaining a dynamo are for the planet’s core to contain “a large (big D) volume of conductive fluid (small η) that is undergoing convection (non-zero v)” (Brain et al. 3). Using the composition of water worlds first proposed by

Leger et al, that is, 1 $M_{\oplus}$ metals, 2 $M_{\oplus}$ silicates, and 3 $M_{\oplus}$ water, and assuming similar ratios of elements as of that in our solar system (supported by Brewer et al), water worlds would have a more than sufficient volume of conductive fluid. However, they likely would not be able to sustain convection. Convection occurs when the total rate of heat dissipating from the core is higher than the rate at which it can diffuse via conduction (Zuluaga et al. 4). This forces the fluid to physically churn to distribute the heat, and that churning creates a magnetic field. In a Leger water world, “subsolidus convection in ice is not even sufficient to remove the heat” generated by decay of radioactive elements in the core, which is a negligible proportion of the total heat that is released (Leger et al. 501; Zuluaga et al. 4). Because water is unable to remove heat via convection or conduction, the temperature of the silicate mantle would slowly rise, lowering the temperature gradient between the inner and outer core, eventually causing convection to cease as the temperature becomes evenly distributed. This would suggest that, all else being equal, water worlds are less likely to host intrinsic magnetic fields.

How would this differ from a gas dwarf of the same size? Gas dwarves would have a larger core and silicate mantle to account for their upper envelope being made of hydrogen and helium gas rather than water. Without the bottleneck of weakly conductive ice to kill the dynamo, they could more plausibly sustain an intrinsic magnetic field. Indeed, it would be expected to be larger than the initial magnetic field of a water world of the same size, as a larger core is the trait most directly correlated to strong active magnetic fields (Glassmeier and Heyner 373). Gas dwarves also need a protective magnetic field to stop their gas envelopes from being irradiated by solar weather. Therefore, if a Sub-Neptune planet is observed with a magnetic field, it is more likely to be a gas dwarf.

In addition to the dynamo, a magnetic field can be generated via induction effect from a saltwater ocean, as is believed to be the case on Europa and Callisto (Stevenson 653). Induced

magnetic fields are created not from a planet's own internal forces, but from interactions between the flow of a conductive component such as saltwater with its host's magnetic field (Zimmer et al. 1). The strength of induced fields thus depends on the field of the host star. Because it varies with the variation of the star, it will be more difficult to detect than a more stable dynamo-generated field. The magnetic field on Europa supported the claim that there was a subsurface ocean when it was first observed (Kuskov & Kronrod 551). The variation of magnetic fields on these satellites, and the effects they induce on their host planets, is analogous to that of planets and stars. Despite being more difficult to detect, they could be studied to better understand how high water content relates to a planet's magnetic field and habitability.

Having shown that a water world is not likely to retain its dynamo-generated magnetic field for long, it is still necessary to investigate how it would affect the planet's formation, and vice versa. When a dynamo stops convecting, the planet's magnetic field does not stop instantly. Instead, it diffuses over time according to the rate of magnetic diffusion (Stevenson 653). For a Sub-Neptune, the time to diffuse would be similar to Earth, which is 10^4 years (Brain et al. 3). The inward migration of a water world, or any planet of a similar size, is also 10^4 years, which is "much shorter than... estimated planetary formation timescales" (Terquem et al. 329). Because both processes happen so quickly, and concurrently with planet formation, they are likely to influence each other. It is possible that the dynamo-generated intrinsic magnetic field would cause star-planet interactions by the time it got to its final orbital radius.

Magnetic fields have significant effects on their planets, particularly their atmospheres, and these effects have historically been ignored when discussing water world habitability. Looking at a water world that has arrived at an orbit within the habitable zone of its host star, the rate of irradiation of whatever atmosphere it has will depend on the type of star it orbits. Most stars in the milky way are M-dwarfs, a type of star that is cooler and smaller than the Sun and is characterized by

“powerful magnetic dynamos and resulting X-ray to UV (X-UV) activity” (Engle 1). Their lower temperature means that the habitable zone, where liquid water can exist, is closer in. Because the magnetic field of water worlds likely lasts at least until they reach this distance, it would interact with the magnetic field of the star. Due to the exchange of angular momentum, it would become tidally locked, and move even closer (Brain et al. 29). The atmospheres of water worlds are thus in triple jeopardy: they face more overall radiation, they have no magnetic field to protect themselves, and they are likely to become tidally locked.

Water world atmospheres would initially be comprised of 90% water, 5% ammonia, and 5% carbon dioxide (Leger et al. 502). Ammonia “is very sensitive to UV,” and around a sun-like planet at an Earth like orbit would be photoevaporated or converted to molecular nitrogen and hydrogen in less than 2 million years (Leger et al. 502). The higher UV radiation of M-dwarf stars would increase that rate. According to Leger et al, hydrogen is generally susceptible to hydrodynamical escape, and Brain et al. determine that a lack of magnetic field likely increases this rate of escape. While this escape would sweep away some of the heavier elements, it would be primarily hydrogen and helium (Leger et al. 502). This would leave higher concentrations of carbon, nitrogen, and oxygen in the atmosphere. During periods of intense radiation such as stellar flares, “production of nitrogen oxides should increase,” compounded by the higher ratio of these molecules, causing a “corresponding decrease in oxide molecules, including ozone” (Brain et al. 27). Ozone protects the surface of Earth from dangerous UV radiation. The surface of water worlds would thus be inhospitable to life as we know it.

There may be other factors that would allow water worlds to be habitable. While the lack of a magnetic field typically leads to higher irradiation of protective molecules in the atmosphere, life could still exist deeper in the water. Water absorbs light energy and harmful radiation the same way an atmosphere does. This would form a goldilocks depth in the ocean where life would be protected

from harmful radiation, while still being close enough to the surface to receive some light energy. In Earth's oceans, UVB radiation levels decrease faster than visible light at all salt concentrations (Dunne and Brown 113). This means that, assuming a similar ocean to Earth, organisms at the correct depth would receive energy in the visible light spectrum without being exposed to UVB radiation. Thus, the lack of ozone in the atmosphere does not preclude the existence of life on a water world.

Supplying nutrients such as iron and phosphorus to this goldilocks depth is the larger problem. Since the oceans of a water world would be up to 100km deep, and temperature and pressure at the bottom would be great, organisms would rely on nutrients being brought up to their habitable level by abiological processes (Leger et al. 503). This great pressure may supply a mechanism to bring those nutrients up. Leger et al. note that due to the low thermal conductivity of high-pressure ice, the only way for heat to escape from the core to the surface would be via "melting of ice and migration of the melt towards the upper ocean" (501). The melting and rising of ice from the silicate mantle boundary could bring up elements necessary for life.

The significance of exoplanet habitability, and detection of life outside of Earth, is one of the core driving forces of exoplanet research. NASA's 30-year astrophysics roadmap lists the possibility of extra-terrestrial life as "one of our oldest and most enduring questions" (Kouveliotou et al. 22). Astrobiologists assume that to be habitable, a planet must have liquid water and an atmosphere containing carbon dioxide, water, and molecular nitrogen (Kopparapu et al. 1). These conditions must also be stable enough for a long enough time for life to emerge. Studying exoplanet formation and evolution, and understanding their common compositions, allows researchers to understand "how special (or not) our home is and how close our nearest neighbours might be" (Jackson).

Detecting the magnetic field of Sub-Neptune exoplanets is just within reach, and confirmation of one would be a worthwhile near-term goal. Possible observations have been noted, such as YZ Ceti b and τ Boötis b, but consensus has not yet been achieved (Pineda et al. 1; Turner et al. 1). This is because these observations rely on seeing how the planet affects the star's magnetic field and measuring that. Confirming that those observations are caused by a planet and not some other force from the star itself require long term predictable periodic observations, which requires “significant investment of observing resources as well as a mature understanding of stellar flare behaviors” (Brain et al. 16).

Developing the technology to measure exoplanetary magnetic fields more easily would not only enhance our ability to find earth-like planets, it would also help differentiate between water worlds and gas dwarfs. The degeneracy between them arises because of them both having the same density. Either a small amount of heavy core with a large amount of medium-weight water, or a larger heavy core with a moderate amount of very light gas. Both result in the same mass and radius measurements using Radial Velocity and Transit methods to measure each. Adding a third measurement, the magnetic obstacle size via measuring star-planet interactions as described by Brain et al, would thoroughly distinguish between the two (16). Because a water world is unlikely to have a magnetic field, its obstacle size would be equal to its radius. The magnetic field of a gas dwarf would extend the obstacle size measurably beyond its radius. Thus, comparing the difference between the two radii would allow researchers to determine the strength of the magnetic field.

Magnetic fields of exoplanets remain a promising frontier of near-term research. The technology to reliably measure them is close to being realized, and upcoming missions like the ARIEL mission and Habitable Worlds Observatory will bring even deeper understanding of exoplanet magnetic fields and formation in general. The more that is learned about exoplanets, the more we must appreciate Earth for its seeming rarity and conduciveness to life. It remains the only

planet amenable to life, and even though technology is advancing rapidly, it is likely that Earth will remain the only planet we have, so we must take care to ensure it remains hospitable to life.

Works Cited

- Barros, S. C. C., et al. “The Young Mini-Neptune HD 207496b That Is Either a Naked Core or on the Verge of Becoming One.” *Astronomy and Astrophysics (Berlin)* [Heidelberg], vol. 673, May 2023, p. A4, <https://doi.org/10.1051/0004-6361/202245741>.
- Brain, David A, et al. *Exoplanet Magnetic Fields*. April 2024, <https://doi.org/10.48550/arxiv.2404.15429>.
- Brewer, John M, and Debra A Fischer. “Erratum: ‘C/O and Mg/Si Ratios of Stars in the Solar Neighborhood’ (2016, ApJ, 831, 20).” *The Astrophysical Journal* [Philadelphia], vol. 840, no. 2, May 2017, p. 121, <https://doi.org/10.3847/1538-4357/aa6d53>.
- Dunne, Richard P, and Barbara E Brown. “Penetration of Solar UVB Radiation in Shallow Tropical Waters and Its Potential Biological Effects on Coral Reefs; Results from the Central Indian Ocean and Andaman Sea.” *Marine Ecology. Progress Series (Halstenbek)*, vol. 144, no. 1/3, 1996, pp. 109–118, <https://doi.org/10.3354/meps144109>.
- Engle, Scott G. “Living with a Red Dwarf: X-Ray, UV, and Ca Ii Activity–Age Relationships of M Dwarfs.” *The Astrophysical Journal* [Philadelphia], vol. 960, no. 1, January 2024, <https://doi.org/10.3847/1538-4357/ad0840>.
- Fulton, Benjamin J, et al. “The California-Kepler Survey. III. A Gap in the Radius Distribution of Small Planets.” *arXiv.Org* [Ithaca], June 2017, <https://doi.org/10.48550/arxiv.1703.10375>.
- Glassmeier, Karl-Heinz, and Heyner, Daniel. “Planetary Magnetic Fields.” *Magnetospheres in the Solar System*, John Wiley & Sons, Inc., 2021, pp. 367-389.
- Jackson, Alan. “Re: Undergrad Research Paper - Request for Interview.” Received by Holly Metzger, 14 April 2026.
- Kouveliotou, Chryssa, et al. *Enduring Quests, Daring Visions - NASA Astrophysics in the Next Three Decades*. NASA, January 2014, <https://doi.org/10.48550/arxiv.1401.3741>.

- Kopparapu, Ravi Kumar, et al. "HABITABLE ZONES AROUND MAIN-SEQUENCE STARS: NEW ESTIMATES." *The Astrophysical Journal* [United States], vol. 765, no. 2, March 2013, pp. 131–16, <https://doi.org/10.1088/0004-637X/765/2/131>.
- Kuskov, O.L., and V.A Kronrod. "Internal Structure of Europa and Callisto." *Icarus (New York, N.Y. 1962)*, vol. 177, no. 2, October 2005, pp. 550–69, <https://doi.org/10.1016/j.icarus.2005.04.014>.
- Léger, A, et al. "A New Family of Planets? 'Ocean-Planets.'" *Icarus (New York, N.Y. 1962)* [San Diego, CA], vol. 169, no. 2, June 2004, pp. 499–504, <https://doi.org/10.1016/j.icarus.2004.01.001>.
- Pineda, J. Sebastian, et al. "Evaluating Star–Planet Interactions with Zeeman Doppler Imaging: Case Study in YZ Ceti." *The Astrophysical Journal* [Philadelphia], vol. 997, no. 2, February 2026, <https://doi.org/10.3847/1538-4357/ae2619>.
- Rogers, James G, et al. "Conclusive Evidence for a Population of Water Worlds around M Dwarfs Remains Elusive." *Astrophysical Journal. Letters* [Austin], vol. 947, no. 1, April 2023, p. L19, <https://doi.org/10.3847/2041-8213/acc86f>.
- Sean Carrol, host. "Natalie Batalha on What We Know and Will Learn About Exoplanets." *Mindscape*, episode 326, Wondery, 25 August 2025, www.preposterousuniverse.com/podcast/2025/08/25/326-natalie-batalha-on-what-we-know-and-will-learn-about-exoplanets/. Accessed 13 Feb. 2026.
- Stevenson, David J. "Planetary Magnetic Fields: Achievements and Prospects." *Space Science Reviews* [Dordrecht], vol. 152, nos. 1–4, May 2010, pp. 651–664, <https://doi.org/10.1007/s11214-009-9572-z>.
- Terquem, Caroline, et al. "Disks, Extrasolar Planets and Migration." *Space Science Reviews* [Dordrecht], vol. 92, nos. 1–2, April 2000, pp. 323–340, <https://doi.org/10.1023/A:1005263717117>.

Turner, Jake D. et al. “The search for radio emission from the exoplanetary systems 55 Cancri, υ Andromedae, and τ Boötis using LOFAR beam-formed observations.” *Astronomy & Astrophysics*, vol. 645, no. A59, 2021, <https://doi.org/10.1051/0004-6361/201937201>.

Zimmer, Christophe, et al. “Subsurface Oceans on Europa and Callisto: Constraints from Galileo Magnetometer Observations.” *Icarus (New York, N.Y. 1962)* [Goddard Space Flight Center], vol. 147, no. 2, October 2000, pp. 329–47, <https://doi.org/10.1006/icar.2000.6456>.

Zuluaga, Jorge I, et al. “THE INFLUENCE OF THERMAL EVOLUTION IN THE MAGNETIC PROTECTION OF TERRESTRIAL PLANETS.” *The Astrophysical Journal* [United States], vol. 770, no. 1, June 2013, pp. 23–23, <https://doi.org/10.1088/0004-637X/770/1/23>.